PR – 1

Deep Learning DL Practical 1 Viva Questions (Based on Your Code)

हा practical:

- Boston Housing Dataset
- Linear Regression
- Deep Neural Network (DNN)
- TensorFlow/Keras

वर based आहे.

External almost हे questions विचारू शकतो 💧

BASIC QUESTIONS

1. What is the aim of this program?

Answer:
Boston Housing Dataset वापरून house price prediction करणं using Deep Learning.

---

2. What is Boston Housing Dataset?

Answer:
House-related features आणि prices असलेला dataset.

House price prediction साठी वापरतात.

---

3. Is this regression or classification?

Answer:
Regression 😄
कारण output continuous value आहे (house price).

---

4. What is target variable?

Answer:
House price (`Price` column).

---

5. Why Deep Learning used?

Answer:
Complex relationships learn करण्यासाठी.

---

6. What is DNN?

Answer:
DNN म्हणजे Deep Neural Network.

Multiple hidden layers असलेला neural network.

---

7. What is Neural Network?

Answer:
Human brain inspired interconnected nodes system.

---

8. What is TensorFlow/Keras?

Answer:
Deep learning framework.

Model build आणि train करण्यासाठी वापरतात.

---

9. Why train-test split used?

Answer:
Model performance unseen data वर check करण्यासाठी.

---

10. Why test_size=0.2 used?

Answer:
80% training आणि 20% testing data साठी.

CODE EXPLANATION QUESTIONS

11. Explain this line:

```
boston_data.rename(columns={'medv':'Price'}, inplace=True)
```

Answer:
`medv` column rename करून `Price` केलं readability साठी.

---

12. Why data preprocessing important?

Answer:
Clean आणि normalized data model better train होण्यासाठी.

---

13. Why StandardScaler used?

```
sc = StandardScaler()
```

Answer:
Data normalize करण्यासाठी.

Features same scale मध्ये आणतो.

---

14. Explain fit_transform().

Answer:
Data वर scaling learn करून transformation apply करतो.

---

15. Explain transform().

Answer:
Already learned scaling test data वर apply करतो.

MODEL QUESTIONS

16. What is Sequential model?

```
model = Sequential()
```

Answer:
Layers sequential order मध्ये add करणारा model.

---

17. Explain Dense layer.

```
Dense(128, activation='relu')
```

Answer:
Fully connected neural network layer.

128 neurons आहेत.

---

18. Why multiple Dense layers used?

Answer:
Complex patterns learn करण्यासाठी.

---

19. Why ReLU activation used?

Answer:
Training fast होतं आणि vanishing gradient problem कमी होतो.

---

20. Why final layer has Dense(1)?

Answer:
Single continuous output (house price prediction) साठी.

COMPILATION QUESTIONS

21. Explain model.compile().

```
model.compile(
    optimizer='adam',
    loss='mean_squared_error',
    metrics=['mae']
)
```

Answer:
Model training configuration define करतो.

---

22. What is Optimizer?

Answer:
Weights optimize करणारा algorithm.

---

23. Why Adam optimizer used?

Answer:
Fast convergence आणि adaptive learning मुळे.

---

24. What is Loss Function?

Answer:
Actual आणि predicted output मधला error measure करतो.

---

25. Why MSE used?

Answer:
Regression problems साठी common loss function आहे.

MSE = \frac{1}{n}\sum (y-\hat{y})^2

---

26. What is MAE?

Answer:
Mean Absolute Error.

Average absolute prediction error.

TRAINING QUESTIONS

27. Explain model.fit().

```
model.fit(
    X_train,
    y_train,
    epochs=100,
    validation_split=0.05
)
```

Answer:
Model training सुरू करतो.

---

28. What is Epoch?

Answer:
Complete dataset वर एक full pass.

---

29. Why epochs=100 used?

Answer:
Model properly learn होण्यासाठी.

---

30. What is validation_split?

Answer:
Training data मधला काही भाग validation साठी वापरतो.

EVALUATION QUESTIONS

31. Explain model.predict().

Answer:
New data वर predictions generate करतो.

---

32. Explain model.evaluate().

Answer:
Model performance measure करतो.

---

33. What is R^2 score?

Answer:
Prediction accuracy measure करणारा metric.

---

34. Difference between MSE and MAE?

MSE	MAE
Squared error	Absolute error
Large errors ला more penalty	Simpler

---

35. Why regression model used?

Answer:
Continuous values predict करायचे आहेत.

TRICK QUESTIONS 🚨

36. What happens without scaling?

Answer:
Training unstable आणि slow होऊ शकतं.

---

37. What is Overfitting?

Answer:
Training data वर good पण new data वर poor performance.

---

38. How prevent overfitting?

Answer:

- More data
- Dropout
- Early stopping
- Regularization

---

39. Why Deep Learning better than simple regression?

Answer:
Complex nonlinear relationships learn करू शकतो.

---

40. Why activation function important?

Answer:
Neural network ला nonlinear learning capability देतो 🔥

PR – 2

Deep Learning DL Practical 2 Viva Questions (CNN / Image Classification)

Bhawa snippet fully visible नाहीये 😱
Pan almost confirm aahe ki हा practical:

- CNN
- Fashion MNIST / Image Classification
- TensorFlow/Keras
  वर based आहे 😄💧

External almost हेच questions विचारणार 💯

BASIC QUESTIONS

1. What is the aim of this program?

Answer:
CNN वापरून image classification perform करणं.

---

2. What is CNN?

Answer:
CNN म्हणजे Convolutional Neural Network.

हे image processing आणि image classification साठी वापरतात.

---

3. Why CNN used for images?

Answer:
CNN automatically important image features detect करतो.

Example:
edges, shapes, textures.

---

4. What is Fashion MNIST Dataset?

Answer:
Clothing images dataset image classification practice साठी वापरतात.

---

5. What is Image Classification?

Answer:
Image कोणत्या category मध्ये आहे हे predict करणं.

---

6. What is TensorFlow/Keras?

Answer:
Deep learning framework model build आणि train करण्यासाठी.

---

7. What is Sequential Model?

```
model = Sequential()
```

Answer:
Layers sequential order मध्ये add करणारा model.

CNN LAYERS QUESTIONS

8. What is Conv2D layer?

```
Conv2D(...)
```

Answer:
Image वर filters apply करून features extract करतो.

---

9. What is Convolution Operation?

Answer:
Filters वापरून image मधले important patterns detect करणं.

---

10. What is Filter/Kernel?

Answer:
Small matrix जी patterns detect करते.

Example:
edges, textures.

---

11. What is MaxPooling2D?

```
MaxPooling2D(...)
```

Answer:
Feature map size reduce करतो आणि computation कमी करतो.

---

12. Why pooling used?

Answer:

- Computation कमी करण्यासाठी
- Important features retain करण्यासाठी

---

13. What is Flatten layer?

```
Flatten()
```

Answer:
Multidimensional image data ला one-dimensional vector मध्ये convert करतो.

---

14. What is Dense layer?

```
Dense(...)
```

Answer:
Fully connected neural network layer.

---

15. Why multiple layers used?

Answer:
Complex image features learn करण्यासाठी.

ACTIVATION QUESTIONS

16. What is ReLU activation?

Answer:
Training fast करतो आणि vanishing gradient problem कमी करतो.

---

17. Why ReLU used?

Answer:
Nonlinearity add करण्यासाठी आणि fast learning साठी.

---

18. What is Softmax activation?

Answer:
Output probabilities generate करतो classification साठी.

---

19. Why softmax in output layer?

Answer:
Multiple classes मधून final prediction देण्यासाठी.

TRAINING QUESTIONS

20. What is Epoch?

Answer:
Complete dataset वर एक full pass.

---

21. What is Batch Size?

Answer:
एका वेळेस process होणाऱ्या samples ची संख्या.

---

22. What is Optimizer?

Answer:
Weights optimize करणारा algorithm.

---

23. Why Adam optimizer used?

Answer:
Fast convergence आणि adaptive learning मुळे.

---

24. What is Loss Function?

Answer:
Prediction error measure करणारी function.

---

25. Which loss function used in classification?

Answer:
Sparse Categorical Crossentropy.

---

26. What is Accuracy?

Answer:
Correct predictions percentage.

Accuracy = \frac{Correct\ Predictions}{Total\ Predictions}

CODE FLOW QUESTIONS

27. Why normalize images?

```
x_train = x_train / 255.0
```

Answer:
Pixel values 0–1 range मध्ये आणण्यासाठी.

Training stable होतं.

---

28. Why reshape images?

Answer:
CNN ला proper image dimensions लागतात.

---

29. Explain model.compile().

Answer:
Optimizer, loss function आणि metrics define करतो.

---

30. Explain model.fit().

Answer:
Model training start करतो.

---

31. Explain model.evaluate().

Answer:
Testing data वर performance measure करतो.

---

32. Explain model.predict().

Answer:
New images वर predictions generate करतो.

IMPORTANT CONCEPT QUESTIONS

33. What is Feature Extraction?

Answer:
Image मधले important characteristics identify करणं.

---

34. What is Overfitting?

Answer:
Training data वर good performance पण unseen data वर poor performance.

---

35. How prevent Overfitting?

Answer:

- Dropout
- More data
- Regularization
- Data augmentation

---

36. What is Dropout?

Answer:
Some neurons temporarily disable करून overfitting कमी करणं.

---

37. What is Padding?

Answer:
Image borders भोवती extra pixels add करणं.

---

38. What is Stride?

Answer:
Filter किती pixels move होईल हे stride ठरवतं.

---

39. Why CNN better than ANN for images?

Answer:
CNN automatic feature extraction करतो.

ANN मध्ये manual features लागतात.

---

40. Applications of CNN?

Answer:

- Face recognition
- Medical imaging
- Self-driving cars
- Object detection
- Image classification 🔥

PR – 3

Deep Learning DL Practical 3 Viva Questions (RNN / Time Series Prediction)

Bhawa snippet visible नाही पण almost confirm आहे हा practical:

- RNN / LSTM
- Time Series Forecasting
- Google Stock Price Dataset
  वर based आहे 😄💧

External almost हे questions विचारणार 💯

BASIC QUESTIONS

1. What is the aim of this program?

Answer:
RNN/LSTM वापरून stock price prediction करणं.

---

2. What is RNN?

Answer:
RNN म्हणजे Recurrent Neural Network.

हे sequential data process करण्यासाठी वापरतात.

---

3. Why RNN suitable for time series?

Answer:
RNN previous information remember करतो त्यामुळे sequence understanding better होतं.

---

4. What is Time Series Forecasting?

Answer:
Historical data वापरून future values predict करणं.

---

5. What is Sequential Data?

Answer:
जिथे order important असतो असा data.

Example:
stock prices, text, speech.

---

6. What is LSTM?

Answer:
LSTM म्हणजे Long Short-Term Memory.

हे advanced RNN आहे जे long-term dependencies better handle करतो.

---

7. Difference between RNN and LSTM?

RNN	LSTM
Short memory	Long memory
Vanishing gradient issue	Better memory handling

---

8. Why stock dataset used?

Answer:
Time-series prediction practice साठी.

---

9. What are timestamps?

Answer:
Date आणि time related information.

---

10. Why normalize data?

```
MinMaxScaler()
```

Answer:
Values same range मध्ये आणण्यासाठी.

Training stable आणि faster होतं.

CODE EXPLANATION QUESTIONS

11. What is MinMaxScaler?

Answer:
Data 0–1 range मध्ये normalize करणारा scaler.

---

12. Why reshape() used?

```
reshape()
```

Answer:
RNN ला 3D input shape लागतो त्यामुळे reshape करतात.

---

13. What is input sequence?

Answer:
Previous values चा sequence ज्यावरून next value predict करतो.

---

14. Explain Sequential model.

```
Sequential()
```

Answer:
Layers sequential order मध्ये add करणारा model.

---

15. What is LSTM layer?

```
LSTM(...)
```

Answer:
Sequence patterns learn करणारी memory-based layer.

---

16. Why Dense layer used?

```
Dense(1)
```

Answer:
Final prediction output generate करण्यासाठी.

---

17. Why Dense(1)?

Answer:
Single continuous value predict करायची आहे.

TRAINING QUESTIONS

18. What is Epoch?

Answer:
Complete dataset वर एक full pass.

---

19. What is Batch Size?

Answer:
एका वेळेस process होणाऱ्या samples ची संख्या.

---

20. What is Optimizer?

Answer:
Weights optimize करणारा algorithm.

---

21. Why Adam optimizer used?

Answer:
Fast convergence आणि adaptive learning मुळे.

---

22. What is Loss Function?

Answer:
Prediction error measure करणारी function.

---

23. Which loss function used?

Answer:
Mean Squared Error (MSE).

MSE = \frac{1}{n}\sum (y-\hat{y})^2

---

24. Explain model.compile().

Answer:
Optimizer आणि loss function define करतो.

---

25. Explain model.fit().

Answer:
Model training सुरू करतो.

---

26. Explain model.predict().

Answer:
Future values predictions generate करतो.

IMPORTANT CONCEPT QUESTIONS

27. What is Hidden State?

Answer:
Previous sequence information store करणारी memory.

---

28. What is Vanishing Gradient Problem?

Answer:
Gradients खूप छोटे होतात त्यामुळे learning difficult होतं.

---

29. Why LSTM better than simple RNN?

Answer:
Long-term dependencies better remember करतो.

---

30. What is Overfitting?

Answer:
Training data वर good पण unseen data वर poor performance.

---

31. How prevent overfitting?

Answer:

- More data
- Dropout
- Early stopping

---

32. Why Deep Learning used here?

Answer:
Complex stock price patterns learn करण्यासाठी.

---

33. Why normalization important?

Answer:
Training stability improve करण्यासाठी.

---

34. Why sequence order important?

Answer:
Previous stock prices पुढच्या prices वर effect करतात.

---

35. Why stock prediction difficult?

Answer:
Stock prices highly dynamic आणि unpredictable असतात.

TRICK QUESTIONS 🚨

36. What happens without normalization?

Answer:
Training unstable आणि slow होऊ शकतं.

---

37. Why RNN remembers previous inputs?

Answer:
Hidden state वापरतो.

---

38. Why LSTM popular?

Answer:
Better memory handling आणि improved sequence learning.

---

39. Why time series data sequential असतो?

Answer:
Previous values future values ला affect करतात.

---

40. Applications of RNN/LSTM?

Answer:

- Stock prediction
- Speech recognition
- NLP
- Weather forecasting
- Text generation

🔥 CODE FLOW EXPLANATION

Step 1 — Dataset Load

👉 Stock dataset load करतात.

Step 2 — Normalization

```
MinMaxScaler()
```

👉 Data normalize करतात.

Step 3 — Sequence Creation

👉 Previous values वापरून sequences तयार करतात.

Step 4 — Reshape

```
reshape()
```

👉 RNN input shape मध्ये convert करतात.

Step 5 — Model Build

```
Sequential()
```

👉 Neural network model तयार करतात.

Step 6 — LSTM Layer

```
LSTM(...)
```

👉 Sequential patterns learn करतो.

Step 7 — Dense Layer

```
Dense(1)
```

👉 Final prediction देतो.

Step 8 — Compile

```
model.compile()
```

👉 Optimizer आणि loss define करतो.

Step 9 — Train

```
model.fit()
```

👉 Model training सुरू करतो.

Step 10 — Predict

```
model.predict()
```

👉 Future stock prices predict करतो 💧